SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 – SOFTWARE TESTING

EXPERIMENT 5

E-COMMERCE APPLICATION – NEGATIVE TEST SCENARIOS AND TEST CASES

Aim:

To write a case study to identify and analyze the requirements of E-Commerce Applications, and prepare negative test scenarios as well as test cases for the application.

1. Problem Description

An e-commerce application enables customers to register, log in, search and filter products, view product details, add available products to a cart, provide a delivery address, pay online and track orders.

The E-Commerce Application is expected to provide reliable, secure and user-friendly services. The testing team verifies the stated requirements using valid and invalid inputs, records actual results, and reports any deviation from the expected behaviour.

2. Functional Requirements

1. Users must register with valid name, email, password and mobile number.
2. Users must log in using valid credentials.
3. Product search must return relevant products.
4. Only available products can be added to the cart.
5. Quantity must be greater than zero and not exceed stock.
6. The cart must calculate item totals, discounts, tax and final amount correctly.
7. Valid delivery details must be accepted.
8. An order is confirmed only after successful payment.

4. Negative Test Scenarios and Test Cases

Negative testing checks whether invalid, incomplete, unauthorized or boundary inputs are safely rejected with a clear message.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
NT01	Register without email	Leave email field blank	Required-field validation is displayed.	Pass
NT02	Register invalid email	Enter user@invalid	Email format is rejected.	Pass
NT03	Login with incorrect password	Correct email and wrong password	Login is denied.	Pass
NT04	Add out-of-stock product	Select a product with zero stock	Product cannot be added or ordered.	Pass
NT05	Enter zero quantity	Set cart quantity to 0	Quantity is rejected.	Pass
NT06	Exceed available stock	Request 10 units when only 5 are available	Quantity is rejected with stock message.	Pass
NT07	Invalid PIN code	Enter alphabetic characters in PIN field	Address is not accepted.	Pass
NT08	Declined payment	Simulate payment decline	Order is not confirmed and failure is displayed.	Pass

5. Detailed Execution of a Failed Test Case

Test Case ID: NT06

Precondition: The application is running in the test environment and the relevant test data is available.

Test Steps: 1. Sign in if required. 2. Open the relevant feature. 3. Perform the action: Request 10 units when only 5 are available. 4. Observe the application response.

Expected Result: Quantity is rejected with stock message.

Actual Result: For demonstration, assume the application accepts the invalid input or does not display the required validation message.

Conclusion: The test is marked as failed for defect-report demonstration. This is an illustrative result, not a claim about a deployed application.

6. Sample Bug Report

Field	Description
Bug ID	BR-005
Linked Test Case	NT06
Severity	High (illustrative)
Priority	High (illustrative)
Bug Title	E-Commerce Application does not correctly validate the condition: Exceed available stock.
Environment	Test environment; browser and operating system should be recorded during actual execution.
Preconditions	The application is running in the test environment and the relevant test data is available.
Steps to Reproduce	1. Sign in if required. 2. Open the relevant feature. 3. Perform the action: Request 10 units when only 5 are available. 4. Observe the application response.
Expected Result	Quantity is rejected with stock message.
Actual Result	For demonstration, assume the application accepts the invalid input or does not display the required validation message.
Impact	Invalid data or an unauthorized operation may be accepted, resulting in incorrect processing, data integrity issues or a poor user experience.
Suggested Fix	Apply validation at both client and server sides, return a clear error message, and add regression tests for the boundary and invalid condition.
Status	Open – illustrative sample.

7. Bug Life Cycle

New → Assigned → In Progress → Fixed → Retesting → Closed. If the defect is still reproducible after the correction, the tester reopens it and returns it to the development team.

Conclusion

The test cases prepared for the E-Commerce Application provide coverage of its key functional requirements. Executing them helps identify functional gaps, improve error handling and increase confidence in the quality of the application.